

Spatiotemporal Modeling of Urban Growth in Bengaluru Using Satellite Data

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Abstract—Urban sprawl is one of the prominent issues that arise in expanding cities and causes loss of vegetation, water resources, and agricultural land. Accurate prediction of urban expansion is an important aspect of sustainable urban planning and design of policies. This paper proposes an approach on urban sprawl prediction of Bengaluru by utilizing satellite images of various time intervals based on machine learning techniques. Satellite images of various time intervals are employed for land use and land cover classification, and thus it becomes easy to detect the areas of urban, vegetation, water, and other land elements on the surface of the image. Based on the past land classification phenomenon of the image, various machine learning models are trained to know the characteristics of urban expansion on the image. Based on this, predictions are done on the expansion of Bengaluru city. It is proven that by utilizing machine learning concepts, it becomes easy to know the phenomenon of urban expansion and get insights on expanding Bengaluru

Index Terms—vegetation indices, urban sprawl, machine learning models, spatiotemporal urban analysis

I. INTRODUCTION

Urban sprawl can be described as uncontrolled and fast growth of urban areas towards surrounding natural areas and rural areas. Urban sprawl is normally associated with the transformation of natural vegetation, agricultural lands, and other open areas into urban areas in order to satisfy the demands of a burgeoning population as well as economic activities. Although urbanization is a normal result of urban development, urban sprawl is normally considered a significant issue in urban planning as a result of environmental degradation and loss of green areas associated with sprawl.

The last couple of years have also seen notable growth in the cities of developing countries. One of these is Bengaluru. This is known to be one of the fastest-growing information technology hubs in the country. Growth has brought along its own share of impact. There is land use change because of the impact of growth. This land use change, in turn, is bringing about degradation of vegetative cover and water bodies and

depletion of natural resources. Study of growth pattern within the city is of great significance.

Remote sensing and geographic information systems offer an effective tool to observe urban expansion by assessing land use and land cover changes on a large scale and for a considerable period of time. Multi-temporal remote sensing images offer the ability to obtain both spectral and spatial data, such as vegetation indices like NDVI, to determine vegetation degradation and urban expansion. However, the use of classical statistical and rule-based approaches is often inefficient and incapable of capturing the non-linear and space-time dependencies of urban expansion phenomenon.

Recent developments in machine learning techniques have greatly enhanced the processing capability for large satellite data by allowing models to automatically identify patterns in the historical observations. Through a combination of land use land cover classification maps, vegetation indices, and spatiotemporal patterns, machine learning models can successfully predict future developments for land expansion for urban use. A machine learning-based approach is introduced here for analyzing and predicting the extent of urban sprawl on multi-temporal satellite images. The technique is designed for accurate modeling of land use patterns for future predictions on urban expansion for informed decision support for sustainable urban planning purposes.

II. LITERATURE REVIEW

Urban sprawl and land use land cover change have been extensively studied using remote sensing, geographic information systems, and spatial modeling techniques. Early research primarily focused on detecting urban expansion through satellite-based land cover classification and spatial indicators such as Shannon's entropy, concentric ring analysis, and landscape metrics. These methods were effective in quantifying the extent and direction of urban growth but were largely descriptive in nature and limited to historical analysis.

Later, studies incorporated Cellular Automata methods or Markov chain models into predictive modeling for the simulation of future urbanization. Despite the improvement in spatial simulation capacity, they tended to employ transition rules or assumptions, which turned out less powerful in representing the complex nonlinearity associated with urbanization. Later global research studies applied machine learning or deep learning methods for the analysis of urbanization. Notably, they exhibited their strengths in land use/cover classification or short-term predictions. Still, they were less interested in long-term predictions.

Bengaluru, because of its rapid growth and environmental issues, has been considered for numerous studies using remote sensing techniques. The application of geostatistical models for analyzing the spatial growth of urbanization for landscape management purposes is cited in [1]. The ecological effects brought about by environmental issues such as the urban heat island effect were addressed in [2] and [6] by establishing a correlation between the growth of built-up areas and land surface temperatures.

The long-term change in the use/cover of the Bengaluru metropolitan area has also been studied using multi-temporal satellite images [3], [4]. Although these papers were successful in demarcating the historical transitions occurring in different land covers, they did not combine machine learning models for forecasting. A machine learning model has also been used for the spatial enlargement analysis [5], but its prime emphasis is on visualization and trends [5].

Other analytical methods dealing with Bengaluru covered, among others, the concentric ring analysis [7], and measurement of entropy-based urban sprawl [9]. These studies effectively quantified spatial dispersion using statistical indicators but without predictive learning. Deep learning-based land use classification was proposed in [8] using Convolutional Neural Networks to achieve high accuracy in classification but was limited to land cover mapping for the present. Analyses like [10] focused on green–blue ratio and impervious surface growth for sustainable planning but did not have forecasting-oriented urban growth models.

Recent works published between 2020 and 2023 [11]– [15] focused on hybrid GIS-based machine learning approaches for urban growth analysis with a focus on model comparisons and historical trend evaluations. More recently, spatiotemporal modeling based on learning algorithms in [16]– [20] have been developed, using deep learning and hybrid architectures such as CNN, LSTM, and CA–deep learning models. These, while showing enhanced capability in modeling spatial complexity, often required large labeled datasets and high computational resources and were rarely validated for long-term forecasting of rapidly growing Indian cities.

A. Limitations of Existing Systems

Although major advancements have been achieved in the analysis of urban sprawl, there are certain limitations to current models. In other words, most of today’s models have been specifically designed to identify urban development trends in

the past but are less equipped to predict future development in urban areas.

Traditional systems or GIS systems use predetermined rules or static thresholds or statistical correlations, which make them less adaptable to complex or nonlinear urban dynamics. However, although machine learning methods have been adopted in recent research studies, they typically focus only on land use/land cover classification and not on overall spatiotemporal predictions related to urban growth.

In addition, there is hardly any existing solution that uses multiple information sources such as vegetation indices and land transition information at the same time in its model formulations. The prediction of long-term results based on multi-temporal satellite images is not feasible, especially in areas such as large cities whose expansion is increasing dramatically every year. Also, there is not much automation in existing solutions regarding their entire processing and prediction tasks.

B. Research Gap and Motivation

On the basis of the literature reviewed, there are some major research gaps which are as yet unattended. The key research gaps are related to analyzing historical urban sprawl, and the question is: *How can future urban expansion be accurately forecasted using multi-temporal satellite images?*

Although land use land cover classification and vegetation indices such as NDVI have been widely used, they are typically used separately. This raises questions such as: *How may land use land cover data and vegetation indices be combined in a unified manner for urban growth prediction using machine learning?*

The second open problem is about the modeling capability, which formulates the question of: *How can machine learning models successfully capture the complex nonlinear spatial and temporal processes of urban expansion from satellite imagery without the predefined, static rules and assumptions?*

In fact, the issues related to scalability and applicability are also not very clear-cut in this respect, thus giving rise to the next question: *How can spatiotemporal models for forecasting be made computationally efficient?*

These research questions inform the methodology that shall be presented in the following section.

III. STUDY AREA AND DATASET

The Bengaluru metropolitan region is one of the fastest-growing metropolitan regions in India and is selected as the case study area for this research. The selected region has a geographical boundary defined by Bengaluru and its surrounding peri-urban regions where considerable land use change has occurred over the last thirty years.

Satellite imagery from the Landsat series is used to study the temporal analysis of urban expansion. Datasets from Landsat 5, Landsat 7, and Landsat 9 satellites are used to provide continuous spatial observations from 1988 to 2024. The satellite imagery offers data in the form of visible spectral (RGB) and infrared spectral data with an infrared band. The

resolution of the data is 30m. This means the dataset is in units of $30 \times 30 \text{ m}^2$.

The earlier imagery (1988–2013) was obtained through Google Earth Engine by creating cloud-free annual composites for limited imagery acquisitions. The recent imagery (post-2014) was acquired for five-day revisit periods, causing cloud contamination in some scenes. Cloud masking and annual compositing were required for creation.

Each year's dataset comprises:

- Normalization of RGB bands to range $[-1, 1]$
- Infrared band
- The derived indices: NDVI, NDWI, and NDBI

The categories of land cover in Bengaluru include four:

- Class 0: Other / Background
- Class 1: Water bodies
- Class 2: Vegetation
- Class 3: Built-up / Urban areas

IV. METHODOLOGY

The proposed framework integrates satellite image preprocessing, spectral index extraction, supervised land cover classification, and spatiotemporal simulation using the CA-Markov model to forecast urban growth.

A. Image Preprocessing

All satellite images are radiometrically corrected for normalization in time across various Landsat satellites. The RGB layers are linearly stretched between -1 and 1 to minimize the impact of varying illumination conditions. Pixels with cloud influence are filtered out using cloud masking techniques, and annual composites are created. All images are clipped to the same spatial extent based on the study area in Bengaluru.

B. Spectral Index Extraction

To improve class separability, three spectral indices are computed for each year:

- Normalized Difference Vegetation Index (NDVI)
- Normalized Difference Water Index (NDWI)
- Normalized Difference Built-up Index (NDBI)

These extracted index layers are stacked with bands of RGB and IR images to generate multi-feature inputs for classification tasks.

C. Training Data Generation and Labeling Strategy

Starting from the year 2014 onward, ground truth labels for classification are generated by creating maps of land use derived from Open Street Maps (OSM) data. In cases where OSM data is unavailable for a particular year, a trend classification method is employed by using images in RGB format and spectrally derived indices. Thus, a combined labeling method is employed to generate a labeled dataset for a period of 1988–2024.

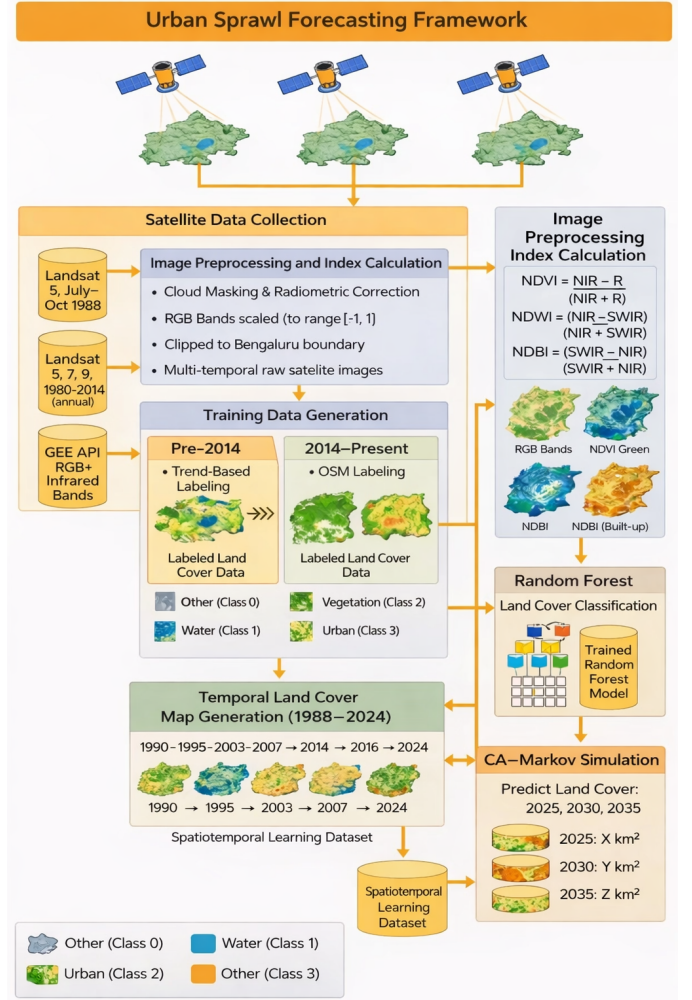


Fig. 1. Overall framework of the proposed urban sprawl forecasting methodology.

D. Random Forest-Based Land Cover Classification

For classifying the pixels using RGB, IR, NDVI, NDWI, and NDBI layers corresponding to one of the four land use classes, a supervised Random Forest classification algorithm is employed. The different parameters required for Random Forest classification are also evaluated, and it has been confirmed that the shallow and conservative models are best suited for land use mapping in all years.

With the selected classifier model, the classification maps of land use-land cover (LULC) are generated for all anchor years. The classification maps of the anchor years provide the required spatiotemporal continuity.

E. Anchor-Year LULC Map Preparation

To ensure consistency for spatiotemporal simulation, all anchor-year LULC maps are reclassified into uniform integer codes and aligned to a common spatial grid with identical coordinate reference system, pixel size, and spatial extent. This step guarantees compatibility of multi-year raster datasets for transition analysis.

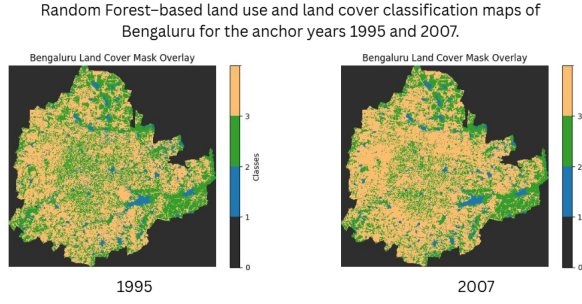


Fig. 2. Random Forest-based land use and land cover classification maps of Bengaluru for the anchor years 1995 and 2007.

OSM-based land use and land cover classification maps of Bengaluru for the years 2016 and 2024.

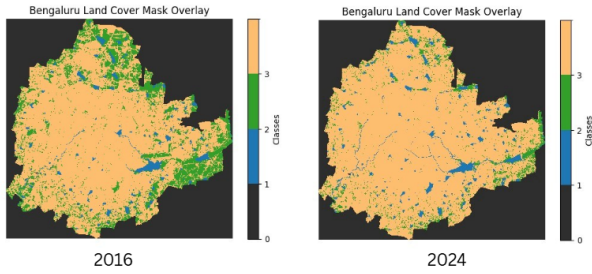


Fig. 3. OSM-based land use and land cover classification maps of Bengaluru for the years 2016 and 2024.

The final selected anchor years are: 1990, 1995, 2003, 2007, 2014, 2016, 2018, 2020, 2022, and 2024.

F. Markov Transition Probability Estimation

Land cover transition matrices are computed using consecutive anchor-year pairs. For each pair, pixel-wise transitions from class i to class j are counted, and transition probabilities are calculated as:

$$P_{ij} = \frac{\text{Number of pixels transitioning from class } i \text{ to } j}{\text{Total number of pixels of class } i}$$

These transition probability matrices provide the time dynamic features of land cover transitions in Bengaluru city.

G. CA-Markov Urban Growth Simulation

Finally, the projection of future urban expansion is performed employing the Cellular Automata-Markov Model. This is because the Markov chain offers transition probabilities concerning time, and the corresponding Cellular Automata part offers influences from the neighboring areas regarding space.

The CA-Markov model uses:

- The most recent LULC map (2024)
- Transition probability matrices from historical anchor years

- Neighborhood spatial rules to control local land conversion

Using this setup, LULC maps are simulated for future target years, specifically 2025 and 2030.

H. Area Estimation

The statistics of urban areas are derived from classification and predicted maps. Because a Landsat satellite image has a resolution of $30 \times 30 \text{ m}^2$, the extent of the urban area is measured by counting how many of its pixels were predicted as urban and calculating their area.

I. Challenges and Limitations

Certain challenges were observed in the preparation of the data and development of the models.

Absence of historical labeled datasets: There are not any ground truth/available LULC benchmarks for the Bengaluru region for preceding decades. Hence, it became mandatory to formulate labeling on trends for images preceding 2014, which might cause limited inconsistencies for class labels in anchor years.

Unavailability of OSM data for past years: Availability of OpenStreetMap land use map layers was limited to the most recent years only. This implied that only post-2014 data had available reliable supervised labels, with prior years having labels obtained only by spectral and temporal consistency.

Limited historical satellite observations: The earlier Landsat satellites had fewer acquisitions per year than their modern counterparts. Therefore, there were limited images available for anchor years, which limited accessibility of cloud-free images.

High cloud intensity in earlier images: Some of the historical images were affected by dense clouds, especially during monsoon season. The clouds were masked and composited in time, but there might still be errors in classification due to the remaining effects of clouds.

Sensor differences across Landsat missions: This made the radiometric normalization of Landsat 5, 7, and 9 very important in order to maintain temporal consistency across the multi-sensor imagery due to variations in spectral response.

Despite these, the adopted pre-processing, normalization, and hybrid labeling strategies enabled construction of a consistent multi-temporal LULC dataset suitable for CA-Markov-based urban growth simulation.

V. EXPERIMENTAL RESULTS

The Random Forest classifier generates stable and accurate LULC maps for all anchor years. The map of classifications shows a continuous expansion of the identified built-up areas away from the center towards the periphery of the city.

Results of the CA-Markov simulation reveal more densification and radial expansion of urban areas around transportation routes. The simulated LULC maps in 2025 and 2030 forecast greater urban area extent at the expense of vegetation cover.

The results of quantitative area analysis do verify the increased area coverage over time, evidencing the potency and

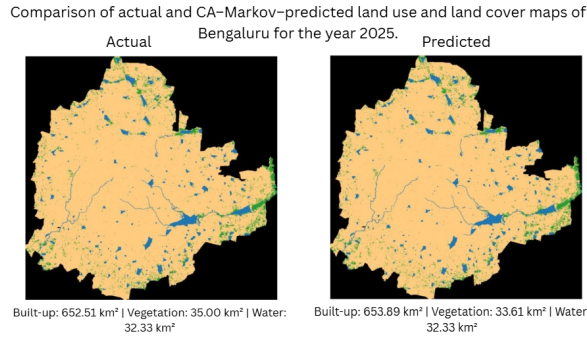


Fig. 4. Comparison of actual and CA-Markov-predicted land use and land cover maps of Bengaluru for the year 2025 with corresponding area statistics.

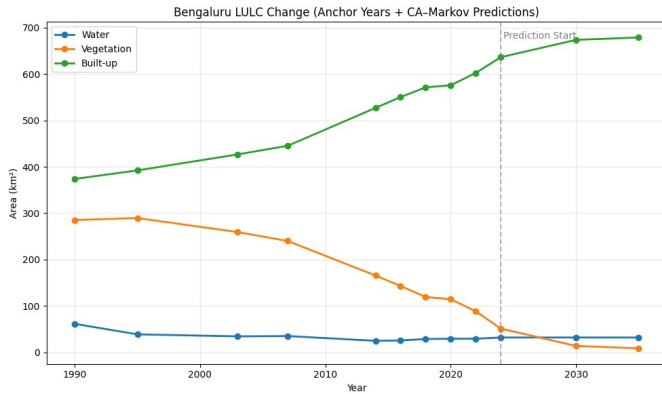


Fig. 5. Temporal variation of land use and land cover areas in Bengaluru using anchor years and CA-Markov predictions.

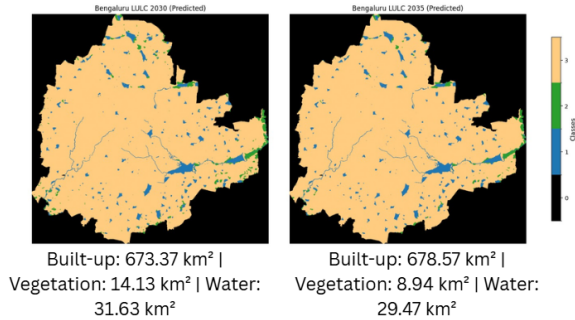


Fig. 6. CA-Markov-predicted land use and land cover maps of Bengaluru for the years 2030 and 2035.

capability of CA-Markov model parameters to portray such spatiotemporal variability for urbanization correctly.

Fig. 5 illustrates the temporal variation of LULC area in Bengaluru from 1990 to 2035. The results show a continuous increase in built-up area accompanied by a steady decline in vegetation cover. The dashed vertical line indicates the start of CA-Markov-based future prediction. This trend confirms sustained urban expansion in the study region.

Fig. 6 presents the CA-Markov simulated LULC maps for the years 2030 and 2035. The predicted maps indicate further densification of built-up regions, particularly along major transportation corridors and peripheral zones, suggesting continued outward urban expansion.

A. Anchor-Year Land Cover Change Analysis

The classified land use and land cover (LULC) maps of the anchor years give the timeline of the land use and land cover changes that occurred in Bengaluru. An anchor year represents an important period during the development of Bengaluru.

1990 and 1995: The early anchor years depict Bengaluru as it was prior to major urbanization. The First City area is mainly built up, with a large number of vegetation areas. Natural landmasses also prevail in the peripheral areas. The water bodies appear quite distinct, showing that the natural river system is less intervened in this era.

2003: By 2003, some visible expansion of the urban area take place. The urbanization process commences to expand along transportation routes, and some evidence of the fragmentation of vegetation in the periphery emerges. It represents a beginning of rapid urban sprawl.

2007: The anchor year of 2007 depicts considerable growth of built-up areas in the city center, along with further extensions to the East/SE directions. There has been a considerable reduction of vegetation cover in the peri-urban areas due to conversion of agricultural/open land to residential patterns.

2014: A major transition phase is noticed during 2014, during which the urban patches increase and become more uniform and extensive. A prominent area is now under the built-up class. Some of the smaller water bodies evident during the previous years decrease or get eliminated, which indicates land encroachment due to land filling.

2016: With the provision for the OpenStreetMap maps available, the level of accuracy of the 2016 classification is considerably ensured. The Urban zones extend very huge towards the paths formed by the newly developed infrastructure, where the vegetation is now less dense, limited only in natural/undeveloped regions.

2018 and 2020: During this phase, there is constant urbanization. This is reflected in the spatial features, where there is infilling within the boundary, resulting in more dense urban agglomerations. The vegetation and water features are still paucity and patchy.

2022 and 2024: The last anchor years show the current extent of Bengaluru as a city. Land use is dominated by buildings, with pockets of vegetation and the remaining lakes. The city has engulfed almost all the peripheral regions, marking the continuation of the process of urban sprawl.

In general, the anchor-year sequence reveals an evident pathway of land transition from vegetation/open areas to built-up areas. Based on these temporal LULC classifications, transition probabilities are estimated to drive the CA-Markov simulation for predicting the future growth of urban areas.

VI. CONCLUSION

In this paper, a hybrid approach has been demonstrated, where LULC classification using Random Forest algorithms has been amalgamated with CA-Markov spatial simulation techniques to accurately predict spatiotemporal changes of urban sprawl of Bengaluru using multi-temporal Landsat satellite images.

The produced urban projection forecasts and area figures give insightful information for urban planners and policymakers. Future research will be directed at the use of higher-resolution data from satellites and innovative spatial constraints to improve forecast accuracy.

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